

Lipids isolated effects on coefficients

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```
getwd()
```

```
[1] "C:/Users/luisr/Documentos/Documents/Documentos - Copia/NanoIA/Topicos_em_Nanotecnologia"
```

In this document, I'm going to calculate the mean of the coefficients of each lipid used in the nanostructure formulations.

Lipids:

- Bacuri
- Açaí
- Pracaxi
- Cacao
- Tucumã
- Cupuaçu
- Pequi
- Manga
- Andiroba
- Murumuru
- Buriti
- Jambu
- Urucum
- Ucuuba

I'm going to apply multiple linear regression to determine how each lipid dictates the **Scheffé's coefficients**.

```
coefData <- read.csv("Tabela de coeficientes NanoIA.csv")
```

Let's fit the linear models

B1

```
#| dev: png  
  
library(magick)
```

Warning: pacote 'magick' foi compilado no R versão 4.5.2

Linking to ImageMagick 6.9.13.29

Enabled features: cairo, freetype, fftw, ghostscript, heic, lcms, pango, raw, rsvg, webp

Disabled features: fontconfig, x11

```
B1 <- lm(  
  b1 ~  
    Bacuri +  
    Açaí +  
    Pracaxi +  
    Cacau +  
    Tucumã +  
    Cupuaçu +  
    Pequi +  
    Manga +  
    Andiroba +  
    Murumuru +  
    Buriti +  
    Jambu +  
    Urucum +  
    Ucuuba,  
  data = coefData  
)
```

B1

Call:

```
lm(formula = b1 ~ Bacuri + Açaí + Pracaxi + Cacau + Tucumã +  
  Cupuaçu + Pequi + Manga + Andiroba + Murumuru + Buriti +  
  Jambu + Urucum + Ucuuba, data = coefData)
```

Coefficients:

(Intercept)	Bacuri	Açaí	Pracaxi	Cacau	Tucumã
3.344019	0.153253	-0.902057	-2.891387	-0.934019	-1.704019
Cupuaçu	Pequi	Manga	Andiroba	Murumuru	Buriti
-1.465981	-2.388038	-0.008517	-3.399617	-1.442057	0.452153
Jambu	Urucum	Ucuuba			
-1.761962	-1.320000	0.509713			

`summary(B1)`

Call:

```
lm(formula = b1 ~ Bacuri + Açaí + Pracaxi + Cacau + Tucumã +  
  Cupuaçu + Pequi + Manga + Andiroba + Murumuru + Buriti +  
  Jambu + Urucum + Ucuuba, data = coefData)
```

Residuals:

1	2	3	4	5	6	7
-2.541e-01	2.541e-01	-2.760e-01	-2.760e-01	2.760e-01	2.012e-16	2.760e-01
8	9	10	11	12	13	14
2.541e-01	-2.541e-01	5.301e-01	-2.541e-01	-2.760e-01	6.245e-17	-2.151e-16
15	16	17				
-1.596e-16	-2.541e-01	2.541e-01				

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.344019	1.950302	1.715	0.229
Bacuri	0.153253	1.315303	0.117	0.918
Açaí	-0.902057	1.172407	-0.769	0.522
Pracaxi	-2.891387	1.435899	-2.014	0.182
Cacau	-0.934019	1.009482	-0.925	0.453
Tucumã	-1.704019	1.460446	-1.167	0.364
Cupuaçu	-1.465981	1.009482	-1.452	0.284
Pequi	-2.388038	1.359601	-1.756	0.221
Manga	-0.008517	1.476584	-0.006	0.996
Andiroba	-3.399617	1.476584	-2.302	0.148

Murumuru	-1.442057	1.172407	-1.230	0.344
Buriti	0.452153	1.315303	0.344	0.764
Jambu	-1.761962	1.359601	-1.296	0.324
Urucum	-1.320000	1.055391	-1.251	0.338
Ucuuba	0.509713	1.435899	0.355	0.757

Residual standard error: 0.7463 on 2 degrees of freedom

Multiple R-squared: 0.9522, Adjusted R-squared: 0.6173

F-statistic: 2.843 on 14 and 2 DF, p-value: 0.2905

```
anova(B1)
```

Analysis of Variance Table

Response: b1

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Bacuri	1	0.8960	0.8960	1.6088	0.33232
Açaí	1	0.6389	0.6389	1.1473	0.39624
Pracaxi	1	0.9648	0.9648	1.7323	0.31872
Cacau	1	0.0167	0.0167	0.0299	0.87858
Tucumã	1	0.2222	0.2222	0.3989	0.59222
Cupuaçu	1	2.6882	2.6882	4.8269	0.15914
Pequi	1	1.1496	1.1496	2.0643	0.28732
Manga	1	1.2509	1.2509	2.2460	0.27270
Andiroba	1	3.3564	3.3564	6.0266	0.13350
Murumuru	1	6.1774	6.1774	11.0919	0.07955
Buriti	1	2.8635	2.8635	5.1416	0.15150
Jambu	1	0.6477	0.6477	1.1630	0.39363
Urucum	1	1.2283	1.2283	2.2055	0.27582
Ucuuba	1	0.0702	0.0702	0.1260	0.75654
Residuals	2	1.1138	0.5569		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

B2

```
lm(
  b2 ~
    Bacuri +
    Açaí +
```

```

    Pracaxi +
    Cacau +
    Tucumã +
    Cupuaçu +
    Pequi +
    Manga +
    Andiroba +
    Murumuru +
    Buriti +
    Jambu +
    Urucum +
    Ucuuba,
  data = coefData
)

```

Call:

```

lm(formula = b2 ~ Bacuri + Açaí + Pracaxi + Cacau + Tucumã +
    Cupuaçu + Pequi + Manga + Andiroba + Murumuru + Buriti +
    Jambu + Urucum + Ucuuba, data = coefData)

```

Coefficients:

(Intercept)	Bacuri	Açaí	Pracaxi	Cacau	Tucumã
0.97157	-4.37140	0.72528	3.02038	-0.67157	-0.05157
Cupuaçu	Pequi	Manga	Andiroba	Murumuru	Buriti
-0.55843	0.41685	-2.29251	1.39149	0.22528	-0.57740
Jambu	Urucum	Ucuuba			
-0.66685	-0.54000	-0.24362			

B3

```

lm(
  b3 ~
    Bacuri +
    Açaí +
    Pracaxi +
    Cacau +
    Tucumã +
    Cupuaçu +
    Pequi +

```

```

Manga +
Andiroba +
Murumuru +
Buriti +
Jambu +
Urucum +
Ucuuba,
data = coefData
)

```

Call:

```

lm(formula = b3 ~ Bacuri + Açaí + Pracaxi + Cacau + Tucumã +
  Cupuaçu + Pequi + Manga + Andiroba + Murumuru + Buriti +
  Jambu + Urucum + Ucuuba, data = coefData)

```

Coefficients:

(Intercept)	Bacuri	Açaí	Pracaxi	Cacau	Tucumã
3.1889	-3.3860	-1.0166	-0.1061	-1.2689	-1.4289
Cupuaçu	Pequi	Manga	Andiroba	Murumuru	Buriti
-2.0811	-2.9477	-2.4496	-0.7326	-0.6166	0.4110
Jambu	Urucum	Ucuuba			
-2.3523	-1.6600	-2.3431			

B12

```

B12 <- lm(
  b12 ~
  Bacuri +
  Açaí +
  Pracaxi +
  Cacau +
  Tucumã +
  Cupuaçu +
  Pequi +
  Manga +
  Andiroba +
  Murumuru +
  Buriti +
  Jambu +

```

```

    Urucum +
    Ucuuba,
    data = coefData
)

```

B12

Call:

```

lm(formula = b12 ~ Bacuri + Açaí + Pracaxi + Cacau + Tucumã +
    Cupuaçu + Pequi + Manga + Andiroba + Murumuru + Buriti +
    Jambu + Urucum + Ucuuba, data = coefData)

```

Coefficients:

(Intercept)	Bacuri	Açaí	Pracaxi	Cacau	Tucumã
0.495745	-1.284043	-0.487234	2.063830	-0.295745	-0.295745
Cupuaçu	Pequi	Manga	Andiroba	Murumuru	Buriti
-0.104255	-4.311489	-3.835106	4.614894	-0.487234	-0.554043
Jambu	Urucum	Ucuuba			
-0.008511	-0.200000	-6.386170			

summary(B12)

Call:

```

lm(formula = b12 ~ Bacuri + Açaí + Pracaxi + Cacau + Tucumã +
    Cupuaçu + Pequi + Manga + Andiroba + Murumuru + Buriti +
    Jambu + Urucum + Ucuuba, data = coefData)

```

Residuals:

1	2	3	4	5	6	7
1.276e+00	-1.276e+00	9.574e-02	9.574e-02	-9.574e-02	-3.331e-16	-9.574e-02
8	9	10	11	12	13	14
-1.276e+00	1.276e+00	-1.371e+00	1.276e+00	9.574e-02	1.665e-16	8.327e-16
15	16	17				
3.886e-16	1.276e+00	-1.276e+00				

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.495745	6.743120	0.074	0.948
Bacuri	-1.284043	4.547627	-0.282	0.804

Açaí	-0.487234	4.053567	-0.120	0.915
Pracaxi	2.063830	4.964586	0.416	0.718
Cacau	-0.295745	3.490259	-0.085	0.940
Tucumã	-0.295745	5.049457	-0.059	0.959
Cupuaçu	-0.104255	3.490259	-0.030	0.979
Pequi	-4.311489	4.700788	-0.917	0.456
Manga	-3.835106	5.105253	-0.751	0.531
Andiroba	4.614894	5.105253	0.904	0.461
Murumuru	-0.487234	4.053567	-0.120	0.915
Buriti	-0.554043	4.547627	-0.122	0.914
Jambu	-0.008511	4.700788	-0.002	0.999
Urucum	-0.200000	3.648987	-0.055	0.961
Ucuuba	-6.386170	4.964586	-1.286	0.327

Residual standard error: 2.58 on 2 degrees of freedom

Multiple R-squared: 0.8089, Adjusted R-squared: -0.5287

F-statistic: 0.6047 on 14 and 2 DF, p-value: 0.7734

B13

```
lm(
  b13~
  Bacuri +
  Açaí +
  Pracaxi +
  Cacau +
  Tucumã +
  Cupuaçu +
  Pequi +
  Manga +
  Andiroba +
  Murumuru +
  Buriti +
  Jambu +
  Urucum +
  Ucuuba,
  data = coefData
)
```


Call:

```
lm(formula = b13 ~ Bacuri + Açaí + Pracaxi + Cacau + Tucumã +  
  Cupuaçu + Pequi + Manga + Andiroba + Murumuru + Buriti +  
  Jambu + Urucum + Ucuuba, data = coefData)
```

Coefficients:

(Intercept)	Bacuri	Açaí	Pracaxi	Cacau	Tucumã
-12.997	7.894	4.641	5.565	6.127	6.127
Cupuaçu	Pequi	Manga	Andiroba	Murumuru	Buriti
7.613	12.794	6.970	6.490	4.641	-1.546
Jambu	Urucum	Ucuuba			
8.356	6.870	6.045			

B23

```
lm(  
  b23~  
  Bacuri +  
  Açaí +  
  Pracaxi +  
  Cacau +  
  Tucumã +  
  Cupuaçu +  
  Pequi +  
  Manga +  
  Andiroba +  
  Murumuru +  
  Buriti +  
  Jambu +  
  Urucum +  
  Ucuuba,  
  data = coefData  
)
```

Call:

```
lm(formula = b23 ~ Bacuri + Açaí + Pracaxi + Cacau + Tucumã +  
  Cupuaçu + Pequi + Manga + Andiroba + Murumuru + Buriti +  
  Jambu + Urucum + Ucuuba, data = coefData)
```

Coefficients:

(Intercept)	Bacuri	Açaí	Pracaxi	Cacau	Tucumã
-8.07979	7.07170	-0.06064	2.07681	3.21979	3.21979
Cupuaçu	Pequi	Manga	Andiroba	Murumuru	Buriti
6.50021	6.72957	4.93426	4.21426	-0.06064	3.84170
Jambu	Urucum	Ucuuba			
8.14043	4.86000	2.79681			

Now, I'm going to apply PCA to the lipids alone.

```
#| warning: False
#| message: FALSE

getwd()
```

```
[1] "C:/Users/luistr/Documents/Documents/Documents - Copia/NanoIA/Topicos_em_Nanotecnologia"
```

```
coefSing <- read.csv("coeficientesSingular.csv")

coefSing <- coefSing[rowSums(is.na(coefSing)) < ncol(coefSing), ]

coefSing <- coefSing[, colSums(is.na(coefSing)) < nrow(coefSing)]

coefSing <- coefSing[!is.na(coefSing$Lipid), ]

coefSing <- coefSing[coefSing$Lipid != "", ]

nrow(coefSing)
```

```
[1] 14
```

```
str(coefSing)
```

```
'data.frame':  14 obs. of  7 variables:
 $ Lipid: chr  "Bacuri" "Açaí" "Pracaxi" "Cacau" ...
 $ b1 : num  0.15 -0.9 -2.89 -0.93 -1.7 -1.46 -2.39 -0.01 -3.4 -1.44 ...
 $ b2 : num  -4.37 0.72 3.02 -0.67 -0.05 -0.56 0.42 -2.29 1.39 0.22 ...
 $ b3 : num  -3.39 -1.02 -0.11 -1.27 -1.43 -2.08 -2.95 -2.45 -0.73 -0.62 ...
 $ b12 : num  -1.28 -0.49 2.06 -0.3 -0.3 -0.1 -4.31 -3.84 4.61 -0.49 ...
 $ b13 : num  7.89 4.64 5.57 6.13 6.13 ...
 $ b23 : num  7.07 -0.06 2.08 3.22 3.22 6.5 6.73 4.93 4.21 -0.06 ...
```

```
head(coefSing)
```

	Lipid	b1	b2	b3	b12	b13	b23
1	Bacuri	0.15	-4.37	-3.39	-1.28	7.89	7.07
2	Açaí	-0.90	0.72	-1.02	-0.49	4.64	-0.06
3	Pracaxi	-2.89	3.02	-0.11	2.06	5.57	2.08
4	Cacau	-0.93	-0.67	-1.27	-0.30	6.13	3.22
5	Tucumã	-1.70	-0.05	-1.43	-0.30	6.13	3.22
6	Cupuaçu	-1.46	-0.56	-2.08	-0.10	7.61	6.50

```
pcaData <- coefSing[, c(
  "b1",
  "b2",
  "b3",
  "b12",
  "b13",
  "b23"
)]

rownames(pcaData) <- coefSing$Lipid

View(coefSing)

pca <- prcomp(
  pcaData,
  retx = TRUE,
  center = TRUE,
  scale. = TRUE
)
pca
```

Standard deviations (1, ..., p=6):

```
[1] 1.73521199 1.36250884 0.90404252 0.52289060 0.18260753 0.09249897
```

Rotation (n x k) = (6 x 6):

	PC1	PC2	PC3	PC4	PC5	PC6
b1	-0.2712017	0.6426442	-0.0401711	0.06491091	-0.450825683	-0.55171284
b2	0.4582196	-0.2657773	0.4486059	0.47889924	-0.531936448	-0.07647958
b3	0.5432664	0.1888530	-0.1187909	0.27098674	0.587351726	-0.48648640
b12	0.3928274	-0.2852268	-0.6405119	-0.39944620	-0.389012736	-0.20781873
b13	-0.3323478	-0.5511309	0.3194968	-0.24224270	0.132220380	-0.63840248

```
b23 -0.3956989 -0.3092019 -0.5202826 0.68903519 0.002151847 -0.04846079
```

```
summary(pca)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6
Standard deviation	1.7352	1.3625	0.9040	0.52289	0.18261	0.09250
Proportion of Variance	0.5018	0.3094	0.1362	0.04557	0.00556	0.00143
Cumulative Proportion	0.5018	0.8112	0.9475	0.99302	0.99857	1.00000

```
library(factoextra)
```

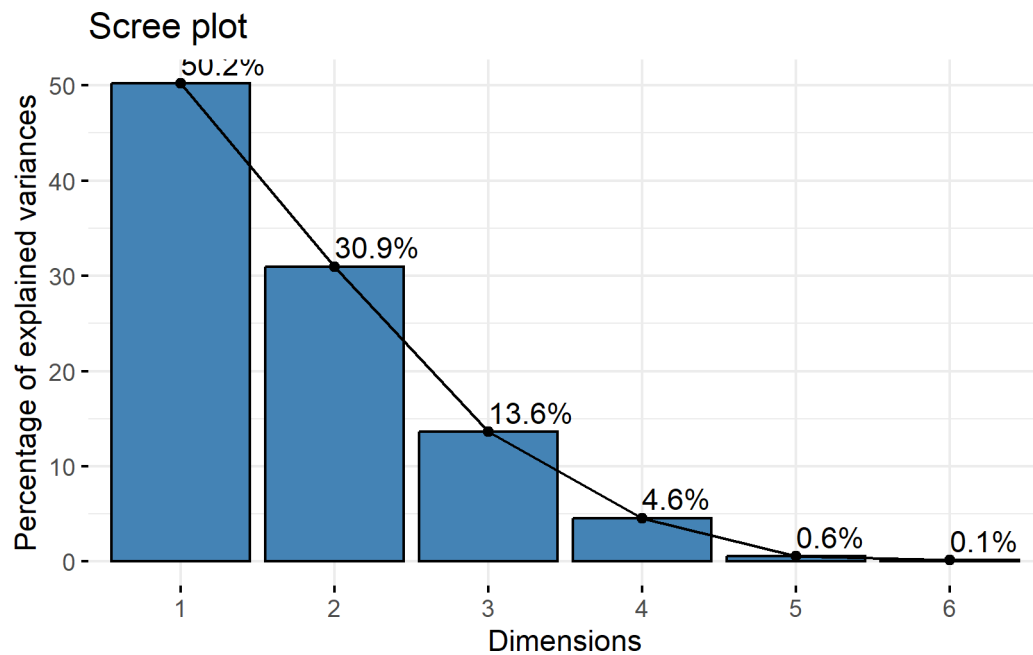
Warning: pacote 'factoextra' foi compilado no R versão 4.5.3

Carregando pacotes exigidos: ggplot2

Welcome to factoextra!

Want to learn more? See two factoextra-related books at <https://www.datanovia.com/en/product>.

```
pcaEigLip <- fviz_eig(pca,
                      barcolor = "black",
                      addlabels = TRUE,
                      ggtheme = theme_minimal(),
                      # parallel = TRUE
                      )
pcaEigLip
```



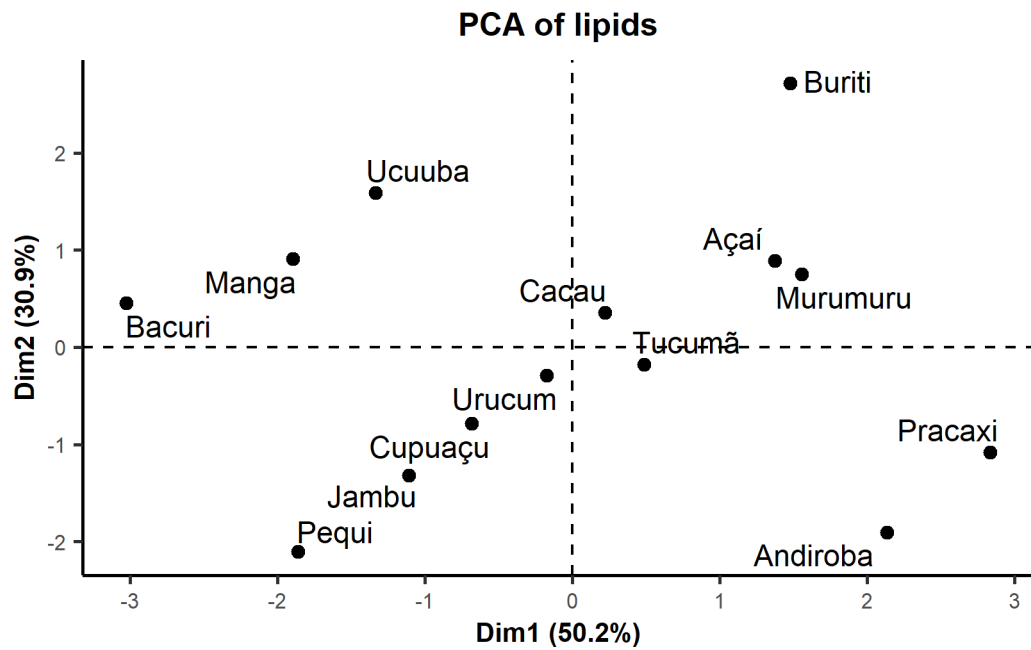
```
ggsave(
  "pcaEigLip.png",
  plot = pcaEigLip,
  width = 18,
  height = 14,
  units = "cm",
  dpi = 300
)

pcaLip <- fviz_pca_ind(
  X = pca,
  repel = TRUE,
  pointsize = 2
) + theme_classic(base_size = 14) +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
      size = 12
    ),
    axis.title = element_text(
      face = "bold",
      size = 10
    ),
  ),
```

```

    axis.text = element_text(size = 8)
  ) +
  labs(
    title = "PCA of lipids"
  )
pcaLip

```



```

ggsave(
  "pcaLip.png",
  plot = pcaLip,
  width = 18,
  height = 14,
  units = "cm",
  dpi = 300
)

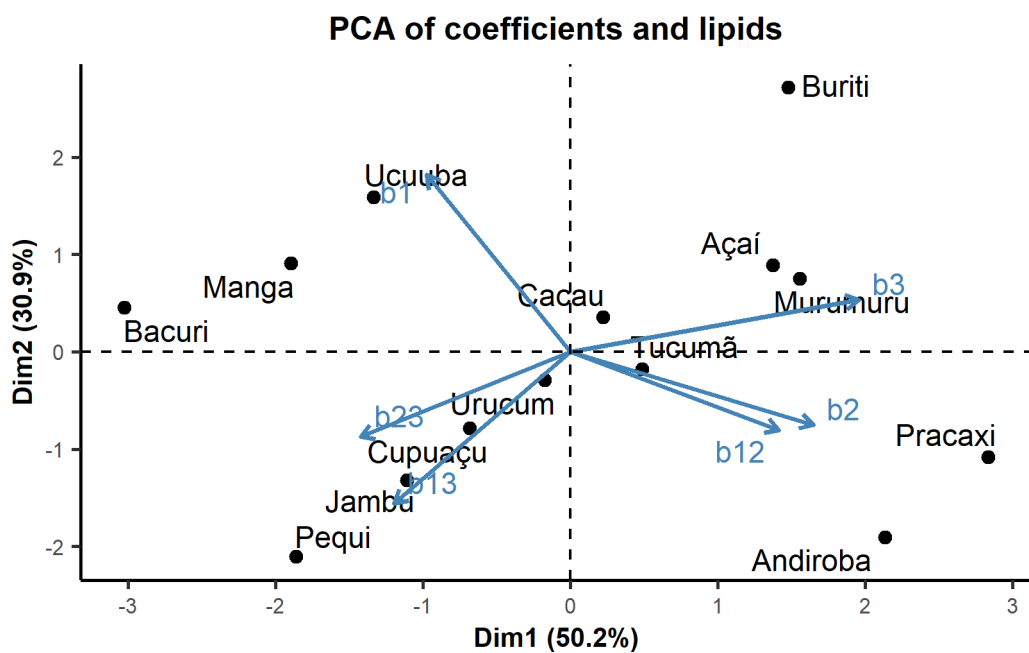
pcaBiplot <- fviz_pca(
  X = pca,
  repel = TRUE,
  #col.var = "#40B5AD",
  #col.ind = "#FA5F55",
  #fill.ind = "contrib",

```

```

pointsize = 2,
arrowsize = 0.8
) + theme_classic(base_size = 14) +
theme(
  plot.title = element_text(
    hjust = 0.5,
    face = "bold",
    size = 12
  ),
  axis.title = element_text(
    face = "bold",
    size = 10
  ),
  axis.text = element_text(size = 8)
) +
labs(
  title = "PCA of coefficients and lipids"
)
pcaBiplot

```



```

ggsave(
  "pcaBiplot.png",
  plot = pcaBiplot,

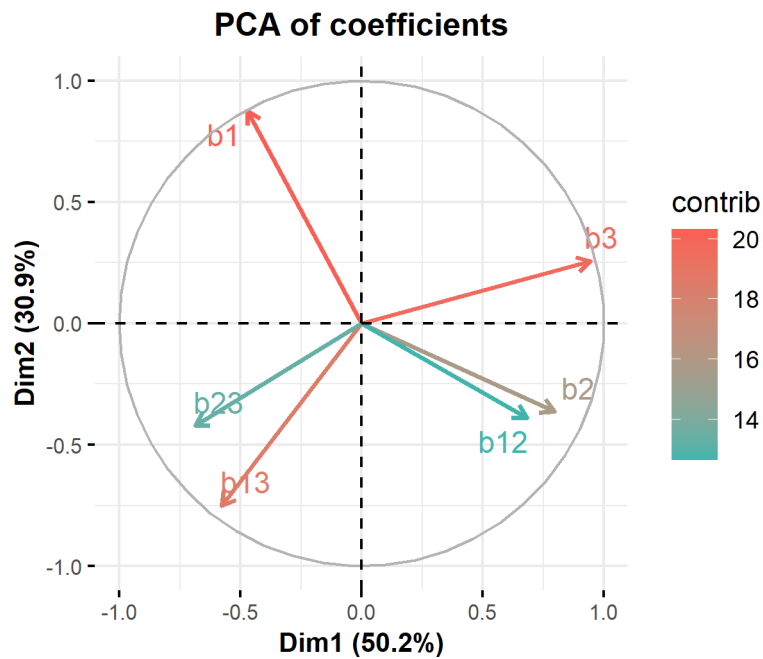
```

```

width = 18,
height = 14,
units = "cm",
dpi = 300
)

pcaVar <- fviz_pca_var(
  X = pca,
  repel = TRUE,
  arrowsize = 0.8,
  col.var = "contrib",
  gradient.cols = c("#40B5AD", "#FA5F55")
) +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
      size = 12
    ),
    axis.title = element_text(
      face = "bold",
      size = 10
    ),
    axis.text = element_text(size = 8)
  ) +
  labs(
    title = "PCA of coefficients"
  )
pcaVar

```

```
ggsave(
  "pcaVarLip.png",
  plot = pcaVar,
  width = 18,
  height = 14,
  units = "cm",
  dpi = 300
)

round(pca$rotation, 3)
```

	PC1	PC2	PC3	PC4	PC5	PC6
b1	-0.271	0.643	-0.040	0.065	-0.451	-0.552
b2	0.458	-0.266	0.449	0.479	-0.532	-0.076
b3	0.543	0.189	-0.119	0.271	0.587	-0.486
b12	0.393	-0.285	-0.641	-0.399	-0.389	-0.208
b13	-0.332	-0.551	0.319	-0.242	0.132	-0.638
b23	-0.396	-0.309	-0.520	0.689	0.002	-0.048

Clustering

```

distMat <- dist(scale(pcaData))

hc <- hclust(
  distMat,
  method = "ward.D2"
)

hc

```

Call:

```
hclust(d = distMat, method = "ward.D2")
```

```

Cluster method   : ward.D2
Distance         : euclidean
Number of objects: 14

```

```

cluster <- cutree(
  hc,
  k = 3
)

cluster

```

Bacuri	Açaí	Pracaxi	Cacau	Tucumã	Cupuaçu	Pequi	Manga
1	2	3	1	1	1	1	1
Andiroba	Murumuru	Buriti	Jambu	Urucum	Ucuuba		
3	2	2	1	1	1		

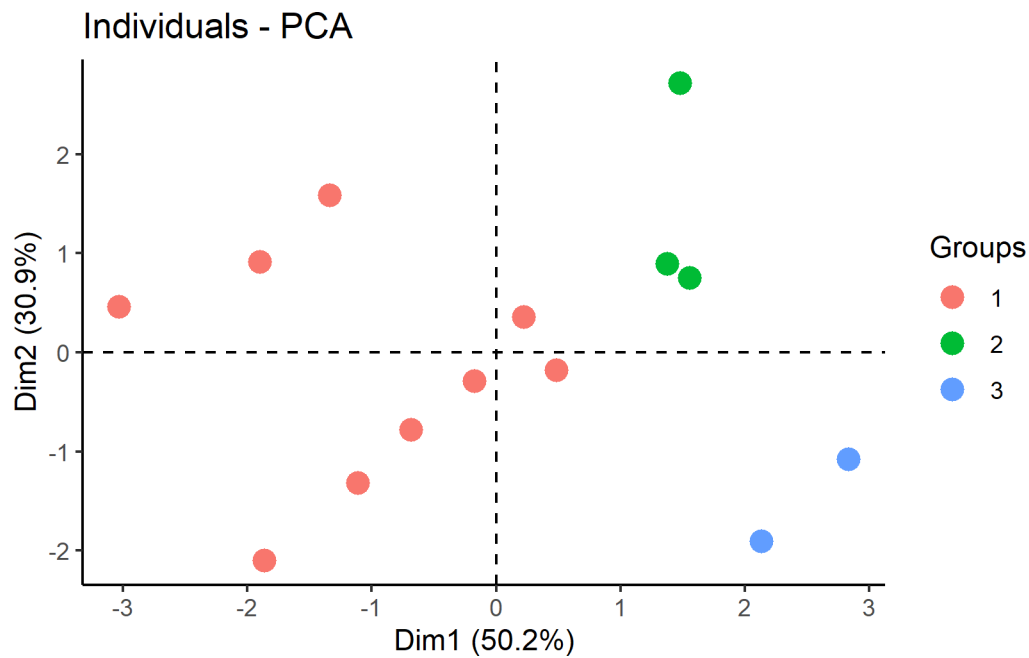
```

pcaHcLipid <- fviz_pca_ind(
  pca,
  geom = "point",
  habillage = factor(cluster),
  pointshape = 16,
  #addEllipses = TRUE,
  fill.ind = "black",
  mean.point = FALSE,
  ellipse.level = 0.95,
  pointsize = 4,
  repel = TRUE
) +

```

```
theme_classic()
```

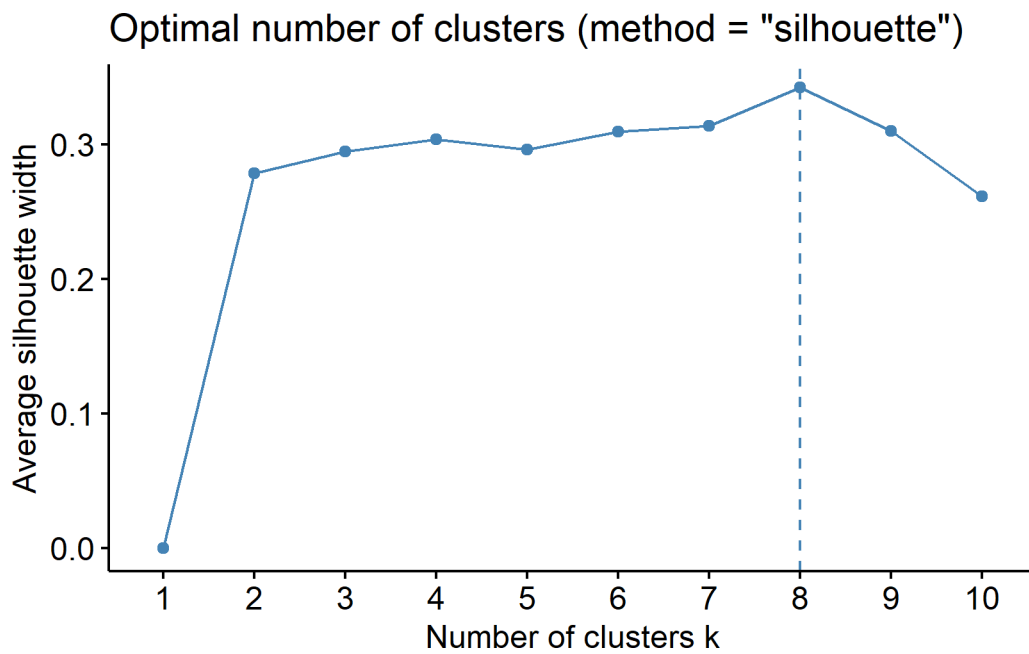
```
pcaHcLipid
```



```
ggsave(  
  "pcaHcLipid.png",  
  plot = pcaHcLipid,  
  width = 18,  
  height = 14,  
  units = "cm",  
  dpi = 300  
)  
  
pcaComp <- fviz_pca_biplot(  
  pca,  
  habillage = factor(cluster),  
  addEllipses = TRUE,  
  ellipse.level = 0.95,  
  pointsize = 4,  
  repel = TRUE  
)  
  
#pcaComp
```

```
#searching the optimal number of clusters
```

```
fviz_nbclust(  
  scale(pcaData),  
  FUNcluster = hcut,  
  method = "silhouette",  
  k.max = 10,  
  print.summary = FALSE  
)
```



```
pcaEclust <- eclust(  
  x = scale(pcaData),  
  FUNcluster = "hclust",  
  k = 4,  
  k.max = 10  
  #graph = TRUE  
)
```

```
pcaEclust
```

Call:

```
stats::hclust(d = x, method = hc_method)
```

```
Cluster method   : ward.D2
Distance         : euclidean
Number of objects: 14
```

```
summary(pcaEclust)
```

	Length	Class	Mode
merge	26	-none-	numeric
height	13	-none-	numeric
order	14	-none-	numeric
labels	14	-none-	character
method	1	-none-	character
call	3	-none-	call
dist.method	1	-none-	character
cluster	14	-none-	numeric
nbclust	1	-none-	numeric
silinfo	3	-none-	list
size	4	-none-	numeric
data	84	-none-	numeric

```
cutree(
  pcaEclust,
  k = 5
)
```

Bacuri	Açaí	Pracaxi	Cacau	Tucumã	Cupuaçu	Pequi	Manga
1	2	3	4	4	4	5	1
Andiroba	Murumuru	Buriti	Jambu	Urucum	Ucuuba		
3	2	2	4	4	1		

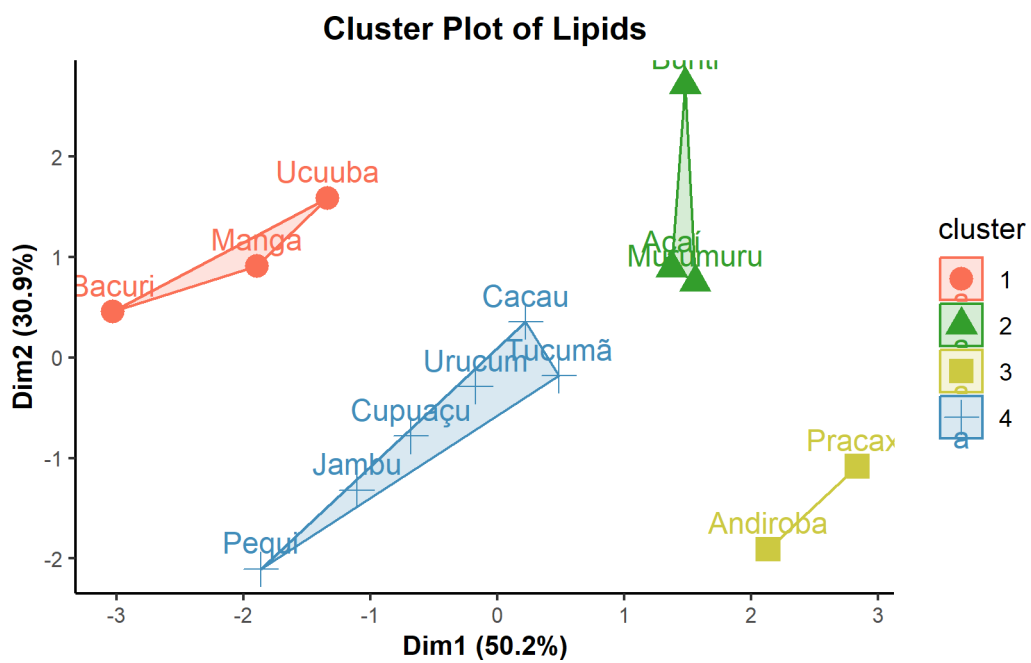
```
pcaCluster <- fviz_cluster(
  pcaEclust,
  #repel = TRUE,
  show.clust.cent = FALSE,
  pointsize = 4,
  palette = c("#FA6F55", "#339E2C", "#CCC941", "#418DBA"),
) + theme_classic() +
theme(
  plot.title = element_text(
```

```

    hjust = 0.5,
    face = "bold",
    size = 12
  ),
  axis.title = element_text(
    face = "bold",
    size = 10
  ),
  axis.text = element_text(size = 8)
) +
labs(
  title = "Cluster Plot of Lipids"
)

```

pcaCluster



```

ggsave(
  "pcaCluster.png",
  plot = pcaCluster,
  width = 18,
  height = 14,
  units = "cm",
  dpi = 300
)

```

```
)

identical(
  pcaEclust$cluster,
  cutree(pcaEclust,k=3)
)
```

[1] FALSE

```
hc <- hclust(
  dist(
    scale(pcaData)
  ),
  method = "ward.D2"
)

cluster1 <- cutree(
  hc,
  k = 3
)

cluster2 <- pcaEclust$cluster

all(cluster1 == cluster2)
```

[1] FALSE

```
cluster2
```

Bacuri	Açaí	Pracaxi	Cacau	Tucumã	Cupuauçu	Pequi	Manga
1	2	3	4	4	4	4	1
Andiroba	Murumuru	Buriti	Jambu	Urucum	Ucuuba		
3	2	2	4	4	1		

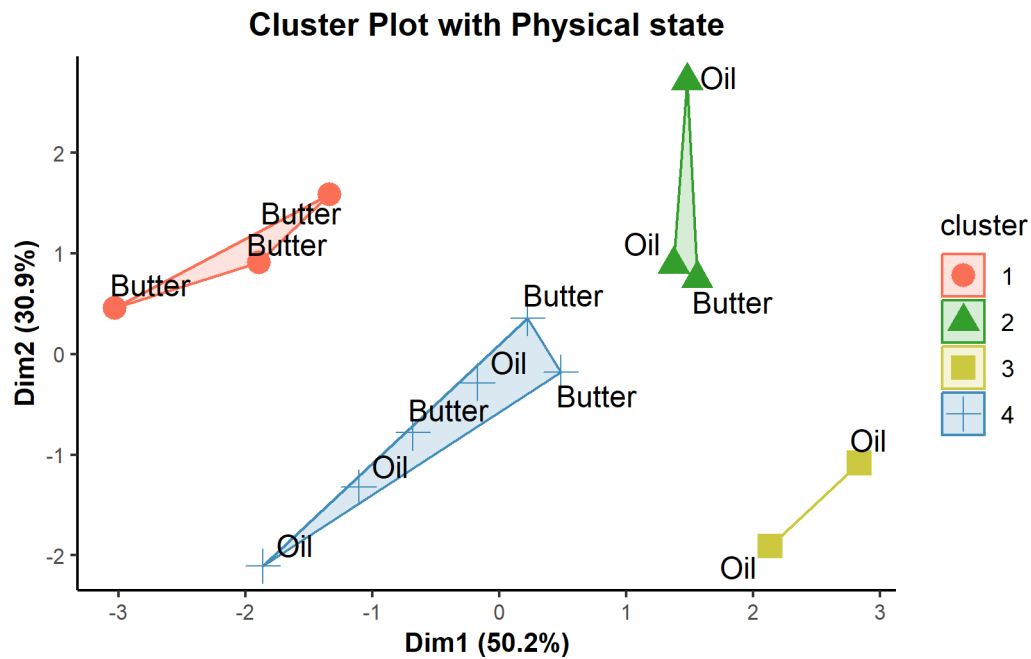
Now, I'll try to plot with the physical state label

```
lipidComp <- read.csv("C:/Users/luisr/Documentos/Documents/Documentos - Copia/NanoIA/Topicos.

library(ggrepel)
```

Warning: pacote 'ggrepel' foi compilado no R versão 4.5.3

```
pcaClusterType <- fviz_cluster(  
  pcaEclust,  
  geom = "point",  
  show.clust.cent = FALSE,  
  pointsize = 4,  
  palette = c("#FA6F55", "#339E2C", "#CCC941", "#418DBA"),  
  
  ) +  
  theme_classic() +  
  geom_text_repel(  
    aes(  
      label = lipidComp$type  
    ),  
    size = 4  
  ) +  
  theme(  
    plot.title = element_text(  
      hjust = 0.5,  
      face = "bold",  
      size = 12  
    ),  
    axis.title = element_text(  
      face = "bold",  
      size = 10  
    ),  
    axis.text = element_text(size = 8)  
  ) +  
  labs(  
    title = "Cluster Plot with Physical state"  
  )  
  
pcaClusterType
```

```
ggsave(
  "pcaClusterType.png",
  plot = pcaClusterType,
  width = 18,
  height = 14,
  units = "cm",
  dpi = 300
)
```

Now I'll plot **fusion point**

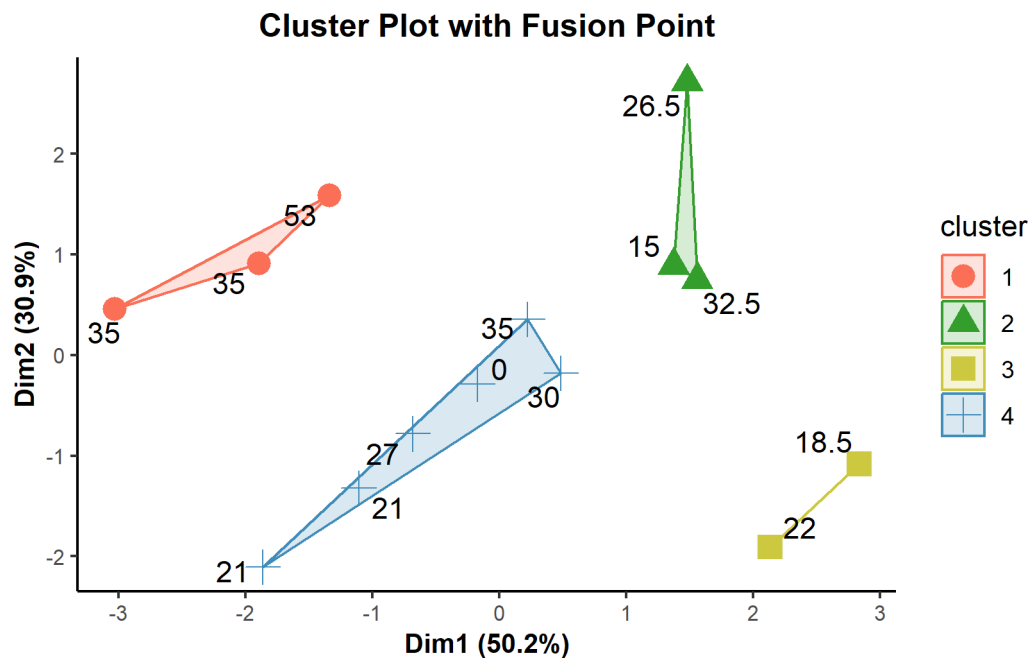
```
pcaClusterFP <- fviz_cluster(
  pcaEclust,
  geom = "point",
  show.clust.cent = FALSE,
  pointsize = 4,
  palette = c("#FA6F55", "#339E2C", "#CCC941", "#418DBA"),
) +
  theme_classic() +
  geom_text_repel(
    aes(
      label = lipidComp$FP
    )
  )
```

```

) +
theme(
  plot.title = element_text(
    hjust = 0.5,
    face = "bold",
    size = 12
  ),
  axis.title = element_text(
    face = "bold",
    size = 10
  ),
  axis.text = element_text(size = 8)
) +
labs(
  title = "Cluster Plot with Fusion Point"
)

pcaClusterFP

```



```

ggsave(
  "pcaClusterFP.png",
  plot = pcaClusterFP,
  width = 18,

```

```

height = 14,
units = "cm",
dpi = 300
)

```

Now, I'll plot with the **percentage of unsaturated fatty acids**

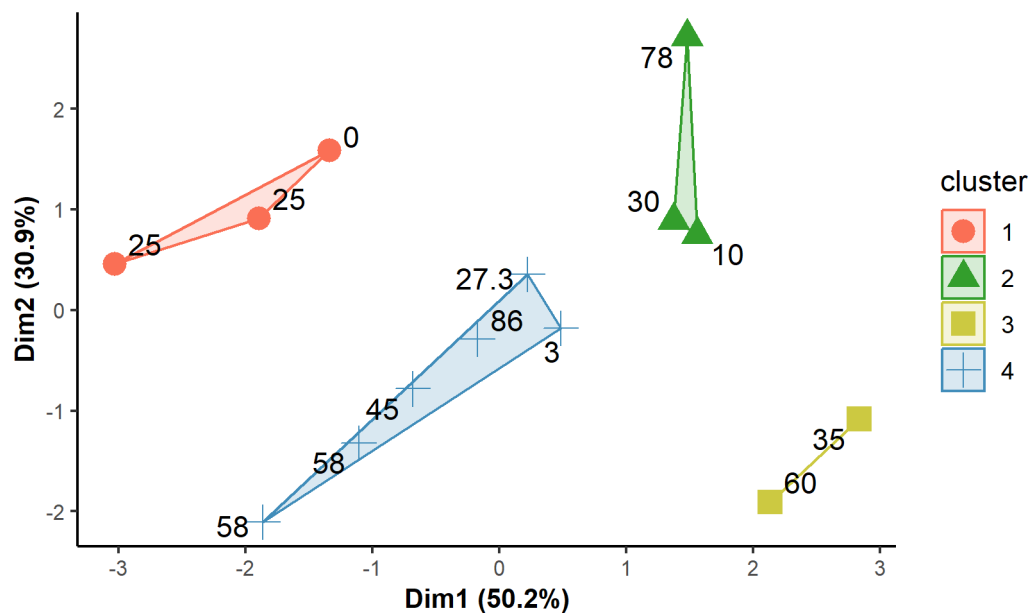
```

pcaClusterUFA <- fviz_cluster(
  pcaEclust,
  geom = "point",
  show.clust.cent = FALSE,
  pointsize = 4,
  palette = c("#FA6F55", "#339E2C", "#CCC941", "#418DBA"),
) +
  theme_classic() +
  geom_text_repel(
    aes(
      label = lipidComp$UFA
    )
  ) +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
      size = 12
    ),
    axis.title = element_text(
      face = "bold",
      size = 10
    ),
    axis.text = element_text(size = 8)
  ) +
  labs(
    title = "Cluster Plot with Percentage of Unsaturated Fatty Acids"
  )

pcaClusterUFA

```

Cluster Plot with Percentage of Unsaturated Fatty Acids



```
ggsave(
  "pcaClusterUFA.png",
  plot = pcaClusterUFA,
  width = 18,
  height = 14,
  units = "cm",
  dpi = 300
)
```

Now with major fatty acid chain

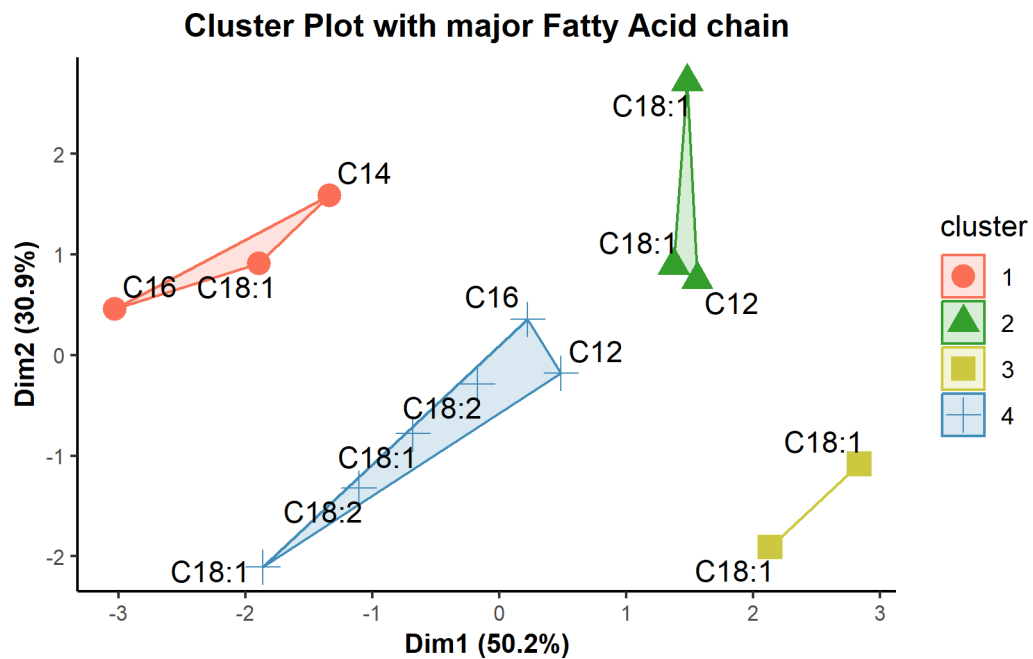
```
pcaClusterFA <- fviz_cluster(
  pcaEclust,
  geom = "point",
  show.clust.cent = FALSE,
  pointsize = 4,
  palette = c("#FA6F55", "#339E2C", "#CCC941", "#418DBA"),
) +
  theme_classic() +
  geom_text_repel(
    aes(
      label = lipidComp$FA
    )
  )
```

```

) +
theme(
  plot.title = element_text(
    hjust = 0.5,
    face = "bold",
    size = 12
  ),
  axis.title = element_text(
    face = "bold",
    size = 10
  ),
  axis.text = element_text(size = 8)
) +
labs(
  title = "Cluster Plot with major Fatty Acid chain"
)

pcaClusterFA

```



```

ggsave(
  "pcaClusterFA.png",
  plot = pcaClusterFA,
  width = 18,

```

```
height = 14,  
units = "cm",  
dpi = 300  
)
```

Dendrogram

```
#library(factoextra)  
library(dendextend)
```

Warning: pacote 'dendextend' foi compilado no R versão 4.5.3

Welcome to dendextend version 1.19.1

Type citation('dendextend') for how to cite the package.

Type browseVignettes(package = 'dendextend') for the package vignette.

The github page is: <https://github.com/talgalili/dendextend/>

Suggestions and bug-reports can be submitted at: <https://github.com/talgalili/dendextend/issues>

You may ask questions at stackoverflow, use the r and dendextend tags:

<https://stackoverflow.com/questions/tagged/dendextend>

To suppress this message use: `suppressPackageStartupMessages(library(dendextend))`

Anexando pacote: 'dendextend'

O seguinte objeto é mascarado por 'package:stats':

cutree

```
dend <- as.dendrogram(hc)  
  
dend <- color_branches(  
  dend,  
  k = 4  
)
```

Carregando namespace exigido: colorspace

```
dev.new()

plot(dend)

abline(
  h = 4,
  col = "black",
  lwd = 2,
  lty = 2
)

class(dend)
```

```
[1] "dendrogram"
```

```
dev.cur()
```

```
pdf
  3
```

```
#png(
#  "dend.png",
#  width = 18,
#  height = 14,
#  units = "cm",
#  res = 300
#)

plot(
  dend,
  main = "Cluster Dendrogram",
  xlab = "",
  sub = "",
  cex = 1.2
)

dev.list()
```

```
png pdf
  2  3
```

```
dev.off()
```

```
png  
  2
```

Heatmap

```
library(pheatmap)
```

Warning: pacote 'pheatmap' foi compilado no R versão 4.5.3

```
pcaDataScale <- scale(pcaData)  
  
heatMap <- pheatmap(  
  pcaDataScale,  
  clustering_method = "ward.D2",  
  border_color = NA,  
  fontsize_row = 12,  
  fontsize_col = 12,  
  main = "Lipid clustering by Scheffé's Coefficients"  
)  
  
class(heatMap)
```

```
[1] "pheatmap"
```

```
heatMap
```


Lipid clustering by Scheffé's Coefficients

